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Federated Learning for Site Aware Chest Radiograph Screening

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ABSTRACT

The shortage of Radiologists is inspiring the development of Deep Learning (DL) based solutions for detecting cardio, thoracic and pulmonary pathologies in Chest radiographs through multi-institutional collaborations. However, sharing the training data across multiple sites is often impossible due to privacy, ownership and technical challenges. Although Federated Learning (FL) has emerged as a solution to this, the large variations in disease prevalence and co-morbidity distributions across the sites may hinder proper training. We propose a DL architecture with a Convolutional Neural Network (CNN) followed by a Graph Neural Network (GNN) to address this issue. The CNN-GNN model is trained by modifying the Federated Averaging algorithm. The CNN weights are shared across all sites to extract robust features while separate GNN models are trained at each site to leverage the local co-morbidity dependencies for multi-label disease classification. The CheXpert dataset is partitioned across five sites to simulate the FL set up. Federated training did not show any significant drop in performance over centralized training. The site-specific GNN models also demonstrated their efficacy in modelling local disease co-occurrence statistics leading to an average area under the ROC curve of 0.79 with a 1.74% improvement.

***Index Terms*—** Chest X-ray, Federated learning, Graph Neural Networks, Convolutional Neural Networks.

1. INTRODUCTION

Chest X-ray radiographs (CXR) are routinely employed in clinics to detect several cardio, thoracic and pulmonary pathologies. The shortage of Radiologists [1] is inspiring the development of deep learning (DL) based screening solutions to reduce the burden of routine screening on Radiologists and help minimize the intra- and inter-expert variations in diagnosis. CXR screening is a *multi-label* classification task where multiple disease classes can be present in the same CXR image. Moreover, many of these classes tend to co-occur, ie., the presence of one class statistically increases (or decreases) the probability of occurrence of other *comorbid* disease classes.

A comprehensive evaluation of different Convolutional Neural Network (CNN) architectures in [2] found the ResNet [3] and DenseNet [4] architectures to be most suitable, with a majority of the existing methods adapting one of these architectures [5, 6, 7, 8] for screening CXR images. Few existing methods have also attempted to model the co-morbidity dependencies among the disease classes by employing conditional Training [9], a Recurrent Neural Network that considers the disease classes to be in a sequential ordering [10] and a way to combine the predictions of an ensemble of CNN architectures using a Graph Neural Network (GNN) [11].

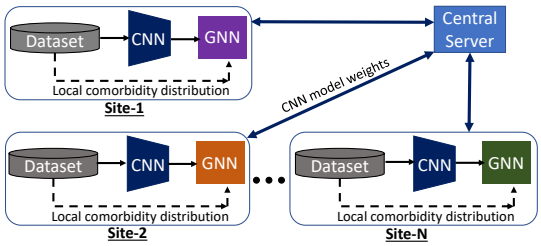

Fig. 1. A sequence of CNN-GNN model is trained across N sites. A global CNN model weight is shared across all sites while separate GNN model weights are learned at each site to capture the local co-morbidity dependencies among the disease classes. The central server periodically receives the local updates of the CNN weights from each site, aggregates them and sends back the global CNN weights to each site using Federated Learning.

DL models require a large training dataset to prevent over-fitting which is often collected through multi-institutional collaborations. The training data is often hosted across multiple sites and cannot be transferred to a central server for training due to patient privacy, data ownership and technical concerns related to the storage and transfer of a large amount of data. Recent work in [12], [13] have explored Federated Learning (FL) as an attempt to address this issue. In FL, a CNN model is trained at each local site and the updated CNN model weights from all sites are periodically aggregated and shared using a central server. Sharing the model weights with the central server instead of the training data is advantageous because the model weights do not contain individual patient-specific information and are relatively small in size thereby requiring lower network bandwidth and storage space in the central server.

However, large variations in the distribution of disease prevalence and comorbidities across geographically distant sites pose additional challenges. In such cases, while training separate CNN models at each site may lead to over-fitting due to the small training sample size at each site, a global

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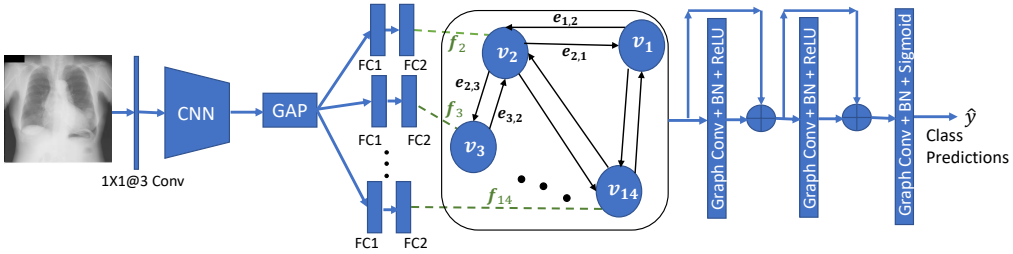

Fig. 2. The proposed CNN-GNN architecture. Separate Fully Connected (FC) layers are employed to obtain different 512-D features for each class. These are used as node features to construct a graph whose edges capture the co-occurrence dependencies between the classes at each site. The graph is provided as input to a Graph Neural Network to obtain the prediction labels for each node. The entire CNN-GNN architecture is trainable in an end-to-end manner.

CNN model trained using FL is unable to capture the local site-specific variations in the data distributions. In order to address this issue, we propose a DL architecture composed of a CNN model followed by a Graph Neural Network (GNN) architecture as depicted in Fig. 1. While the CNN model weights are learned through FL and shared across the N sites to learn robust site-independent features, a separate GNN is constructed and fine-tuned at each site to leverage the local site-specific disease prevalence and comorbidity statistics for disease classification.

2. METHOD

The proposed CNN-GNN architecture is depicted in Fig. 2. The grayscale CXR images are first mapped to 3 channels using a convolutional layer with three 1×1 filters followed by Batch Normalization (BN) and ReLU activation. This layer is followed by a standard CNN architecture which is initialized with the ImageNet [14] pre-trained weights. The fully connected (FC) layers of the CNN for classification are replaced by a Global Average Pooling (GAP) operation to map the final feature channels into a 1-D feature vector. Unlike the standard CNN architectures which share the same FC layer features for all classes, we employ separate Blocks of FC layers for each class after the GAP operation to represent each class i with a different feature vector $\boldsymbol{f}_i \in \mathbb{R}^{512}$. Each block consists of 2 FC layers, each of which has 512 neurons and employ a BN followed by the ReLU activation function.

Graph Construction: The output features $\boldsymbol{f}_i$ from the CNN are employed in the construction of a Graph $G(V, E)$. Each node $\boldsymbol{v}_i \in V$ denotes the ith disease class and is represented using the associated feature vector $\boldsymbol{f}_i$ obtained from the CNN. Since the CNN weights are shared across all sites, the node features $\boldsymbol{f}_i$ are only dependent on the input CXR image and independent of its source site. Directed edges are defined between each pair of nodes $(\boldsymbol{v}_i, \boldsymbol{v}_j) \in E$ to model the co-occurrence dependencies between the disease classes. The edge between $(\boldsymbol{v}_i, \boldsymbol{v}_j)$ is characterized using an edge feature $\boldsymbol{e}_{i,j} \in \mathbb{R}^4$ which is defined as $\boldsymbol{e}_{i,j} = (P_{0,0}, P_{0,1}, P_{1,0}, P_{1,1})$, where $P_{u,v} = P(i = u, j = v)$ is the fraction of samples in the training set for which the Ground Truth label for the ith class is u and the jth class is v. At each site, the edge features are defined using the local training data resulting in a *different graph structure for each site*. Thus, all CXR images from the same source site will share the same edge features but have different node features.

GNN Architecture: The GNN architecture accepts the node features from the output of the CNN and the edge features as input to predict a score $\hat{\boldsymbol{y}}_i \in [0, 1]$ for each $\boldsymbol{v}_i$ indicating the presence of the corresponding disease class. A weighted binary Cross-Entropy loss is defined for each $\hat{\boldsymbol{y}}_i \in [0, 1]$ independently without a $Softmax$ operation since multiple disease classes can co-occur. The loss function can be backpropagated through the node features thereby allowing an end-to-end training of the entire CNN-GNN architecture. While the node features are learned during training by updating the CNN model weights, the edge features act as an external input to the GNN and used to condition the GNN layers for each local site. The edge features are not updated during training.

As depicted in Fig. 2, the proposed architecture has three edge-conditioned Graph Convolution Layers (GCL) [15]. The first two layers map 512-D input node features to transformed 512-D output node features using a GCL with BN and ReLU activation. The last GCL layer maps the input 512-D node features to a 1-D prediction score using the Sigmoid activation. Additionally, the first two GCL layers also employ residual connections to facilitate better gradient backpropagation during the end-to-end training of the CNN-GNN architecture by avoiding vanishing gradients. The edge-conditioned GCL operation [15] can be mathematically defined as

$$\boldsymbol{f}_i^{out} = g\left(\boldsymbol{W}.\boldsymbol{f}_{(i)}^{inp} + \frac{1}{\mathcal{N}_i}\sum_{j \in \mathcal{N}_i} \boldsymbol{H}(\boldsymbol{e}_{j,i}).\boldsymbol{f}_i^{inp} + \boldsymbol{b}\right). \quad (1)$$

In a GCL, first a learnable basis matrix $\boldsymbol{W}$ is applied to the input node feature vectors $\boldsymbol{f}_i^{inp}$ for each $\boldsymbol{v}_i$ to obtain the transformed node feature $\boldsymbol{W}.\boldsymbol{f}_i^{inp}$. Next, the contributions from all it's neighboring nodes $\boldsymbol{v}_j$ are averaged and added to the transformed node feature along with the bias vector $\boldsymbol{b}$

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Algorithm 1 Federated Averaging with site specific GNNs. **S** is the set of sites.

Server executes:
Initialize $\phi_C^{(0)}$ at Server and $\psi_k^{(0)}$ at each site $k \in \mathbf{S}$.
for each round $t = 1, 2...T$ **do**
  for each site $k \in \mathbf{S}$ **do**
    Send $\phi_C^{(t-1)}$ to site k
    Site k **executes:**
    Freeze $\phi_C^{(t-1)}$
    $\psi_{k,ft}^{(t)} \leftarrow local_train(k, \phi_C^{(t-1)}, \psi_k^{(t-1)})$
    // Fine-tune the GNN weights
    $val_k \leftarrow validation_metric(k, \phi_C^{(t-1)}, \psi_k^{(t-1)})$
    // Compute average AUC on local validation set
    Unfreeze $\phi_C^{(t-1)}$
    $\phi_k^{(t)}$,$\psi_k^{(t)} \leftarrow local_train(k, \phi_C^{(t-1)}, \psi_{k,ft}^{(t)})$
    // End-to-end full model training
    Store $\psi_k^{(t)}$ locally and **Send** $\phi_k^{(t)}$, val_k to server
  end for
  Use $\sum_k val_k$ to monitor performance and save model checkpoints
  $\phi_C^{(t)} = \sum_k \lambda_k.\phi_k^{(t)}$ // update global CNN weights
end for

and the activation function $g(.)$ is applied to obtain the output feature vector $\boldsymbol{f}_i^{out}$. The contribution from each neighboring vertex $\boldsymbol{v}_j$ is computed by transforming its input node feature $\boldsymbol{f}_j^{inp}$ using a learnable basis matrix $\boldsymbol{H}(\boldsymbol{e_{j,i}})$ which is conditioned on the edge feature $\boldsymbol{e}_{j,i}$. In our architecture, $\boldsymbol{H}(\boldsymbol{e}_{j,i})$ is modeled using a FC layer with a $tanh()$ activation which takes the 4-D edge feature $\boldsymbol{e}_{j,i}$ as input and outputs a vector which is reshaped to obtain the basis matrix $\boldsymbol{H}(\boldsymbol{e}_{j,i})$.

Federated Training: We assume the multi-center setting described in Fig. 1. Since the proposed CNN-GNN architecture is trainable in an end-to-end manner, Federated Averaging [16] can be used to learn a single CNN and GNN model weight that is shared across all sites. However, since our aim is to only learn a shared CNN model while training separate local GNN models at each site, we modified the Federated Averaging as depicted in Algorithm 1. A local copy of the CNN-GNN architecture is created at each site k. The training proceeds over T communication rounds during which the central server stores the global CNN weights $\phi_C^{(t)}$ and each site k maintains the local GNN weights $\psi_k^{(t)}$.

At each communication round t, the central server sends the current global CNN weight $\phi_C^{(t-1)}$ to each local site where the local CNN-GNN model is trained in two steps. First, the CNN weight $\phi_C^{(t-1)}$ is frozen and the local GNN weight $\psi_k^{(t-1)}$ is updated through backpropagation for a few batch updates to obtain $\psi_{k,ft}^{(t)}$. Next, the CNN weights are unfrozen and the entire CNN-GNN architecture is fine-tuned in an end-to-end manner to obtain the updated local CNN weight $\phi_k^{(t)}$ and the GNN weight $\psi_k^{(t)}$. The site k saves $\psi_k^{(t)}$ locally and returns $\phi_k^{(t)}$ back to the central server. Once the central server has received the locally trained CNN weights from all sites, it updates the global CNN weight as the weighted average, $\phi_C^{(t)} = \sum_k \lambda_k.\phi_k^{(t)}$ where the weight λ_k is inversely proportional to the number of training samples at site k. The average validation performance across all sites is monitored during each round of training and the optimal set of CNN and local GNN weights is selected as the final model.

3. EXPERIMENTAL RESULTS

Dataset: The proposed method has been evaluated on the publicly available CheXpert dataset [5] which consists of 223,414 CXR images with the Ground Truth (GT) labels for fourteen disease classes. In our experiments, we simulated the multi-center environment by dividing the entire dataset into 5 sites and ensuring that during training, each site only has access to its local data. The data at each site is further sub-divided into training, test and validation sets. The distribution of the disease classes and comorbidities varied significantly across the 5 sites to reflect a real-world setting. There is a high occurrence of cardiac diseases in Site-1, while Site-2 has a higher prevalence of infectious pulmonary diseases as indicated by the presence of Consolidation and pneumonia. Site-3 and Site-4 have a high prevalence of other pulmonary conditions while Site-5 has many healthy and Fracture cases. The details of the data distribution at each site and a csv file with the actual data partition are available in the Supplementary Material [1].

Implementation Details: The proposed architecture is implemented in Python using the Pytorch 1.4.0 and Pytorch Geometric [2] libraries. Similar to [5, 11], the CXR images are resized to 320×320 and whitened to have a zero mean and unit variance. During training, random horizontal flips, crops and rotations within a range of $\pm 10°$ are employed for data augmentation. The model is trained for 40 communication rounds. In each round, each site is trained independently by first fine-tuning the GNN for 1500 batch updates with a batch size of 16 while keeping the global CNN weights frozen. At this time, the global CNN weights are shared across all sites while the GNN weights are fine-tuned specific to each site. Next, an end-to-end training of the entire CNN-GNN model is performed for 3000 batch updates to obtain the weight updates for the CNN for the next round of training. The Adam optimizer [17] with a learning rate of 10^{-4} is employed.

Results: Based on the baseline performance comparison of different CNN architectures for CXR screening in [2], we selected the ResNet-18 [3] CNN architecture for our experiments. In order to assess the impact of FL and the local GNN models, the classification performance of three different variants of the proposed architecture is reported in Table 1 for each site.

[1] available online at `www.bit.ly/Supplementary_FL_GNN`
[2] `https://pytorch-geometric.readthedocs.io/`

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Table 1. Area under the ROC (AUROC) curves for the 14 disease classes for each site using ResNet-18. The average AUROC across all the classes is presented in the last column

	No Finding	Enl. Cardiom.	Cardio-megaly	Lung Opacity	Lung Lesion	Edema	Consoli-dation	Pneum-onia	Atelec-tasis	Pneumo-thorax	Pleural Effusion	Pleural Other	Fracture	Support Devices	Average
Site-1															
CNN(G)	0.9197	0.6120	0.8107	0.7436	0.7842	0.8027	0.6768	0.6865	0.6768	0.8546	0.8720	0.8417	0.7553	0.8850	0.7801
CNN(FL)	0.8133	0.6697	0.7466	0.7235	0.7688	0.7935	0.6590	0.6849	0.6543	0.7928	0.8327	0.8275	0.7356	0.8319	0.7524
CNN(FL)+GNN	0.9074	0.7442	0.7866	0.7312	0.7742	0.7956	0.6435	0.7011	0.6442	0.8064	0.8425	0.8332	0.7226	0.8461	0.7699
Site-2															
CNN(G)	0.8024	0.6919	0.7921	0.768	0.7886	0.8291	0.7835	0.7124	0.7056	0.8510	0.8930	0.7954	0.7591	0.8913	0.7902
CNN(FL)	0.8456	0.8039	0.8142	0.7471	0.7527	0.8235	0.7786	0.6995	0.6954	0.8383	0.8625	0.7733	0.7351	0.8545	0.7874
CNN(FL)+GNN	0.8680	0.7946	0.8038	0.7623	0.7487	0.8233	0.7847	0.7425	0.6909	0.8614	0.8661	0.8016	0.7222	0.8550	0.7947
Site-3															
CNN(G)	0.8752	0.6938	0.8211	0.6359	0.7691	0.8272	0.7123	0.6917	0.6683	0.8561	0.8491	0.8249	0.7687	0.8767	0.7764
CNN(FL)	0.9017	0.8060	0.8430	0.6260	0.7742	0.8233	0.7111	0.6935	0.6681	0.8271	0.8005	0.8223	0.7363	0.8379	0.7765
CNN(FL)+GNN	0.9223	0.8301	0.8376	0.6440	0.7772	0.8234	0.7298	0.7621	0.6720	0.8136	0.8140	0.8212	0.7333	0.8380	0.7870
Site-4															
CNN(G)	0.8132	0.6801	0.7818	0.6358	0.7728	0.8306	0.7687	0.7408	0.6827	0.8609	0.8275	0.782	0.8084	0.8682	0.7753
CNN(FL)	0.8029	0.8414	0.8363	0.6253	0.7537	0.8264	0.7007	0.7430	0.6772	0.8265	0.7861	0.7585	0.7946	0.8263	0.7714
CNN(FL)+GNN	0.8443	0.8717	0.8499	0.6770	0.7467	0.8233	0.7754	0.7429	0.6743	0.8290	0.7962	0.7958	0.7979	0.8261	0.7893
Site-5															
CNN(G)	0.8719	0.6820	0.7706	0.8603	0.6794	0.8880	0.8099	0.7374	0.7588	0.9069	0.9197	0.7189	0.7927	0.9219	0.8085
CNN(FL)	0.7807	0.7479	0.7618	0.8445	0.6346	0.8829	0.8111	0.8267	0.7325	0.8910	0.8988	0.6813	0.7544	0.8905	0.7956
CNN(FL)+GNN	0.8610	0.7563	0.7204	0.8514	0.7001	0.8805	0.8077	0.8193	0.7495	0.7915	0.9047	0.8360	0.7495	0.9155	0.8102

CNN(FL)+GNN denotes the proposed CNN-GNN architecture trained using FL with Algorithm 1. The *CNN(G)* variant only employs the CNN architecture by replacing the GNN with an FC layer with sigmoid activation to obtain the disease class predictions. *CNN(G)* is trained *without* FL by collecting the training data from all the sites to a central server. Although such centralized training may be infeasible in a real-world scenario, it provides an upper bound of the expected performance of the model when trained using FL. While *CNN(FL)+GNN* only needs 74.4 MB of CNN weights to be transferred to the central server from each site during each round of federated training, the *CNN(G)* requires 4.4 GB of training data (after image resizing) to be transferred to the central server first, requiring more storage space and network bandwidth. The third variant *CNN(FL)* uses the same architecture as *CNN(G)* but is trained in a federated manner using Federated Averaging [16].

Comparing the average Area under the ROC (AUROC) curve across the fourteen classes reported in the last column of Table 1, we find *CNN(G)* to perform slightly better than *CNN(FL)* with an average AUROC of 0.7861 for *CNN(G)* and 0.7767 for *CNN(FL)* across the five sites. Thus, FL is a viable option to train DL models for CXR screening in a multi-center study when the training data cannot be shared across sites and it does not lead to a significant drop in performance over centralized training.

Again, comparing the average AUROC across the last column of Table 1, the proposed *CNN(FL)+GNN* shows an improvement of $\frac{0.7902-0.7767}{0.7767} \times 100 = 1.74\%$ over *CNN(FL)* across the five sites, thereby demonstrating the efficacy of the GNN models in modelling the local disease comorbidity distributions at each site. The *CNN(FL)+GNN* is especially useful for the co-morbid disease classes. In site-1, enlarged cardiomediastinum and cardiomegaly are two cardiac disorders that tend to co-occur and *CNN(FL)+GNN* shows $\frac{0.7442-0.6697}{0.6697} \times 100 = 11.12\%$ and $\frac{0.7866-0.7466}{0.7466} \times 100 = 5.35\%$ improvement in performance for these two classes respectively over *CNN(FL)*.

4. CONCLUSION

In this work, we propose a DL architecture with a CNN model followed by a GNN for screening CXR images. Although the entire CNN-GNN architecture is trained in an end-to-end manner, we modified the Federated Averaging algorithm to ensure that while the CNN is jointly trained on data across multiple sites, separate GNNs are trained on the local data at each site. This ensures that the CNN with a larger number of learnable parameters is able to learn robust, site-independent features while the GNN models with few learnable parameters are able to leverage the local disease prevalence and comorbidity distributions at each site for multi-label disease classification. The proposed method can be easily adapted to a new site using a small training set to fine-tune the local GNN model while freezing the pre-trained CNN model weights. Our method achieved an average AUC of 0.79 for 14 disease classes across 5 data sites with large variations in their data distribution. The DL architectures learned in a federated manner did not lead to a significant drop in performance over centralized training. Thus, FL is a promising option to train models for CXR screening in a collaborative multi-center setting without sharing data across the sites. The local GNN models specific to each site led to an overall average improvement in AUROC of 1.74% with larger improvements for comorbidities such as cardiomegaly and enlarged cardiomediastimum. Extensive evaluation of the proposed method on real-world multi-site datasets and incorporation of clinical priors in defining the graph structure for the disease classes are promising directions for future work.

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5. ACKNOWLEDGMENTS

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6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by the CheXpert dataset [5] from the Stanford ML Group . Ethical approval was not required as confirmed by the license attached with the open access data.

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